



EAST WEST UNIVERSITY

LAB REPORT

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Course Name: Electronic Circuits

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Experiment Title: To study the performance of a BJT switch.

Objectives:

1. Understand the fundamental operation of a bipolar junction transistor (BJT) and explore its role in digital switching circuits.
2. Explore the concept of using a transistor as a switch, which forms the Foundation for various digital circuits.
3. Analyze the impact of varying input signals on the switching behavior of the BJT to evaluate its efficiency, speed, and performance in practical applications.

Theory:

A BJT is a three-terminal semiconductor device with two p-n junctions formed by sandwiching either p-type or n-type semiconductor material between a pair of opposite-type semiconductors. The three terminals of a BJT are:

Base (B): Controls the current flow between the other two terminals.

Collector: Collects current from the emitter,

Emitter (E): Emits current into the collector. BJTs can amplify or magnify signals and are commonly used as amplifiers, filters, rectifiers, oscillators, and switches 1.

BJT as a Switch:

When biased in the saturation and cut-off regions, a BJT can function as a switch.

Cut-off Region:

In the cut-off region, no current flows between the collector and emitter. The transistor is effectively "off."

Saturation Region:

In the saturation region, the transistor is fully conducting, allowing current to flow between the collector and emitter.

The transistor is effectively "on."

Using a BJT as a Switch:

Suppose we want to control a lamp using a BJT. We position the collector and emitter where the switch contacts were. By applying the correct biasing voltage, we can turn the lamp on or off. The base current controls the much larger collector current, acting as a current-controlled switch. NPN and PNP transistors can both be used for this purpose.

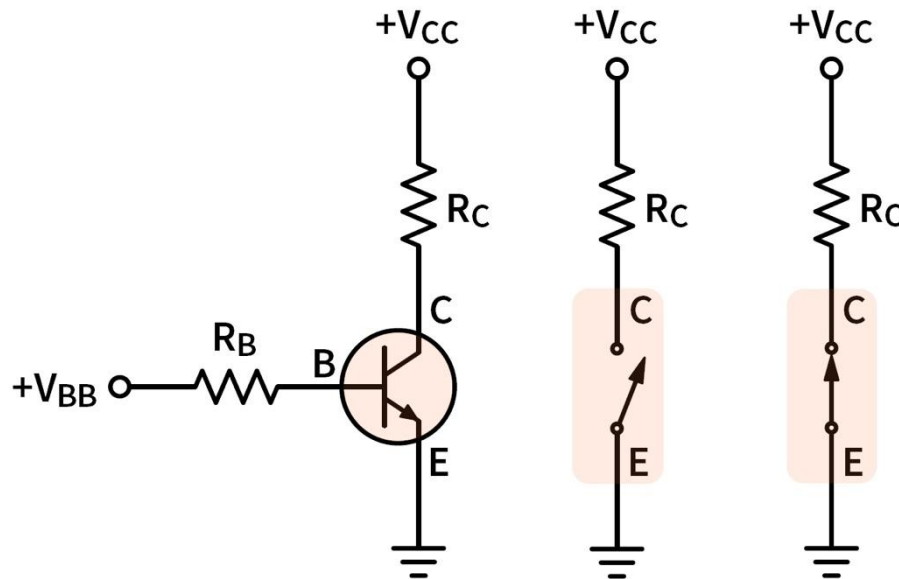


Figure 1: BJT as a Resistive Switch

Apparatus Required:

1. Transistor: 2N2222
2. Resistor $1k\Omega$, $10k\Omega$
3. Breadboard
4. Connecting wires
5. Multimeter
6. Power supply
7. Ammeter
8. LED

Working Diagrams:

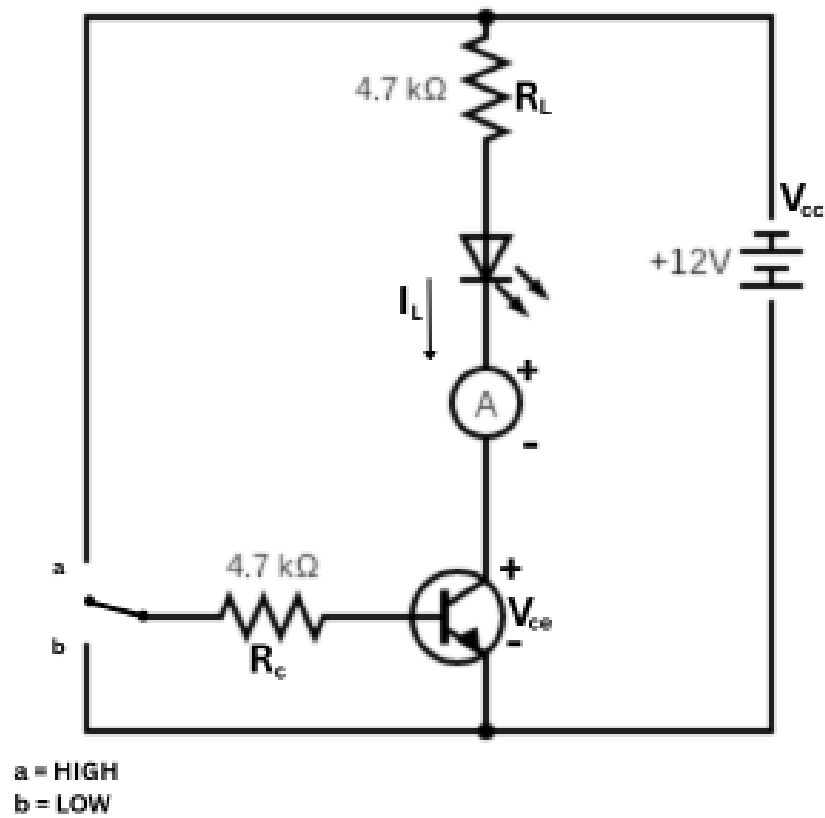


Figure 2: BJT as a Switch

Data Table:

No. of Observation	V_{in} (volt)	State of the LED	I_L (mA)	V_L (volt)	V_{CE} (volt)
1	HIGH	OFF	0	0	11.9
2	LOW	ON	2.2	11.9	0.004

Discussion:

We implemented an N-P-N BJT switch using the provided material. Let's observe the switching behavior. From the Saturation Region, when the base-emitter junction is forward-biased, the transistor enters saturation. Current flows from the collector to the emitter, turning the switch "on.". In the reverse-biased state, the transistor is in cutoff. No current flows, turning the switch "off." That is the Cutoff Region.

We can see that, there are many advantages of BJT Switching. Such that, it can response fast with low voltage drop and have the high current handling capability with simple circuitry. In this experiment, proper base current is crucial and heat dissipation during saturation. Also, Voltage ratings and load characteristics are matter for considerations. There are some applications systems, just like lighting control, motor control, and audio amplification, that are used by the BJT switch.

For future improvements, exploring P-N-P transistors can provide insights into complementary switching behavior. Additionally, implementing more complex circuits, such as H-bridges or multi-stage amplifiers, can expand the scope of the experiment and allow for a deeper understanding of practical applications of BJT switching.